



ANISOTROPIC MATERIALS

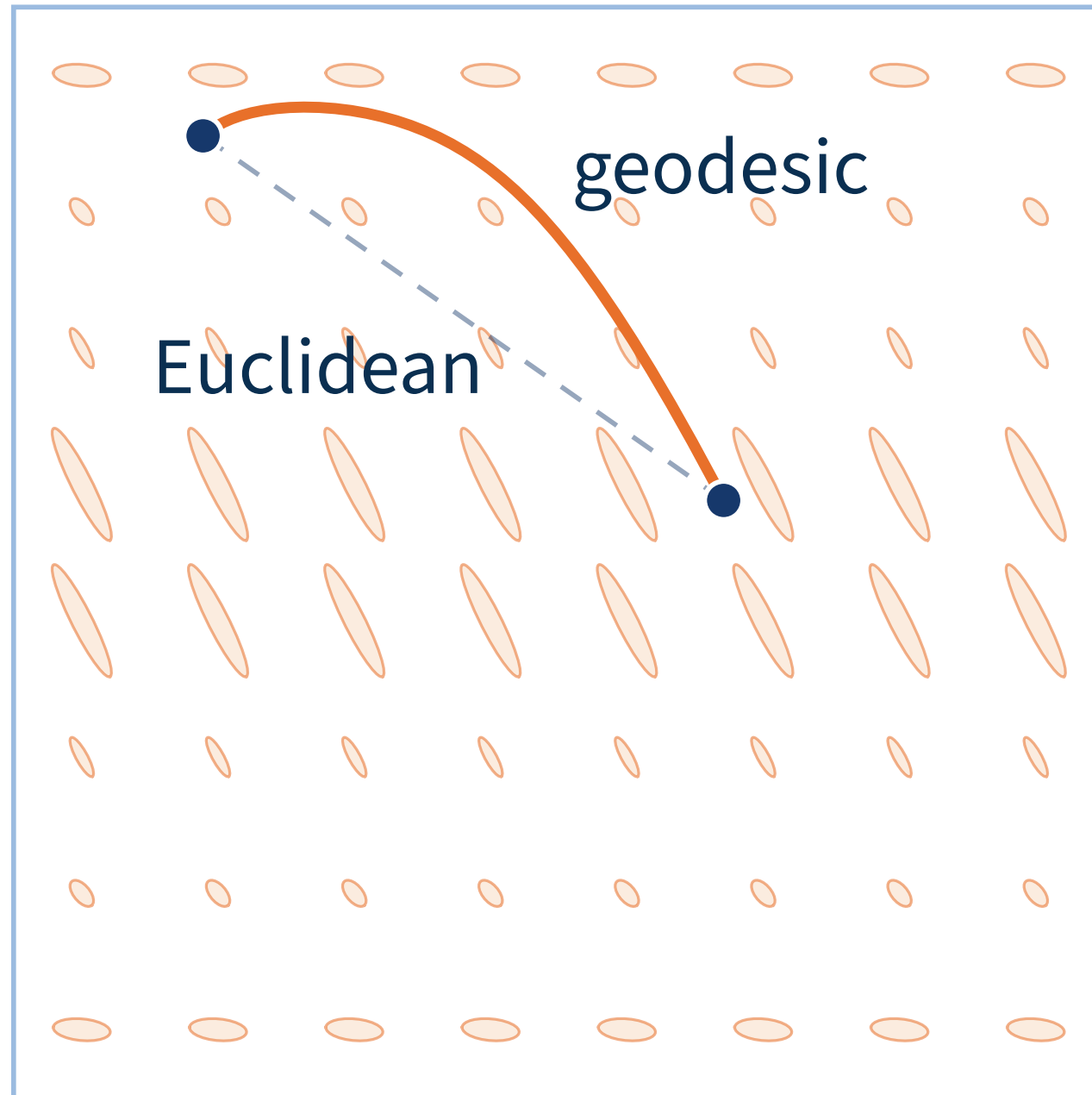
Distance now depends on *where* you are in configuration space

— some directions cost more to move through than others.

TaskSpace



Configuration space



pullback

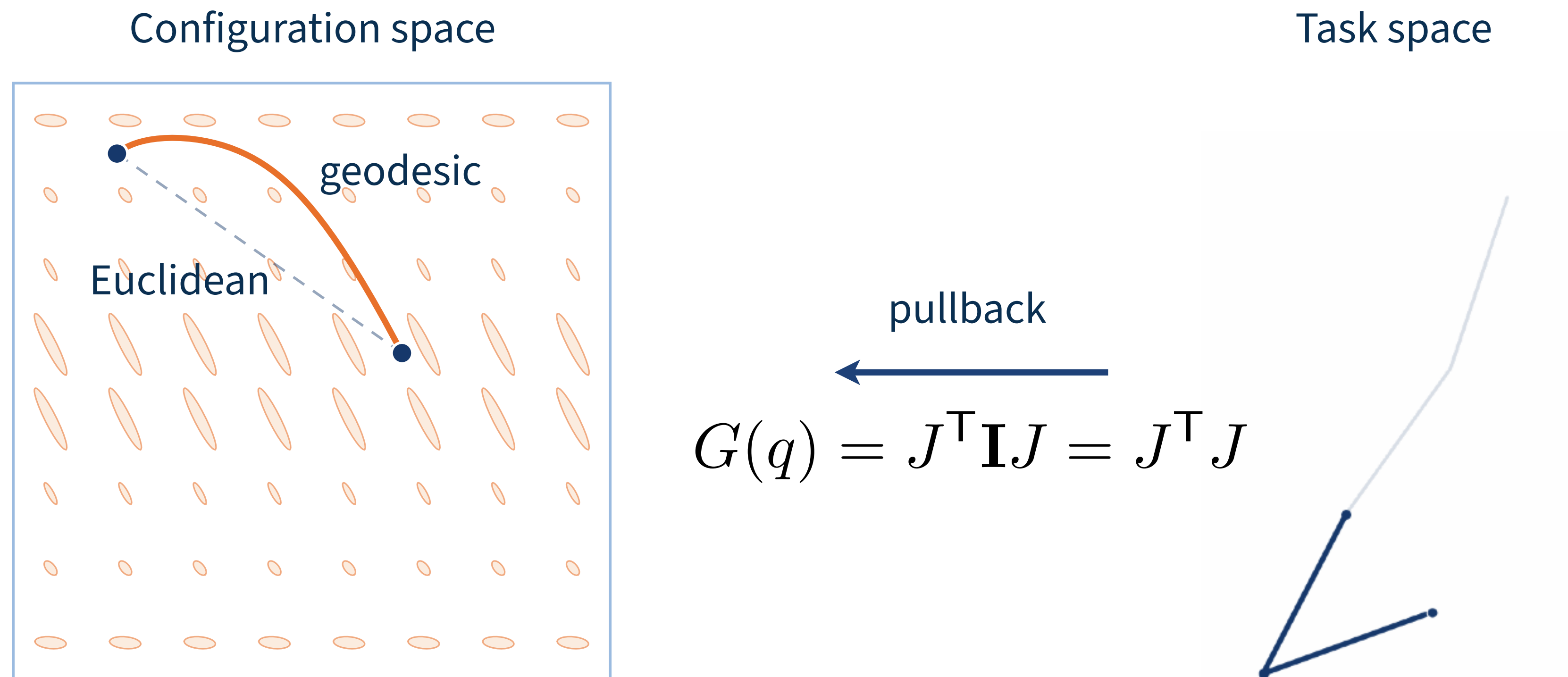
$$G(q) = J^T \mathbf{I} J = J^T J$$

Pull the task-space metric back into configuration space
— a straight line in task space becomes a *geodesic* in configuration space





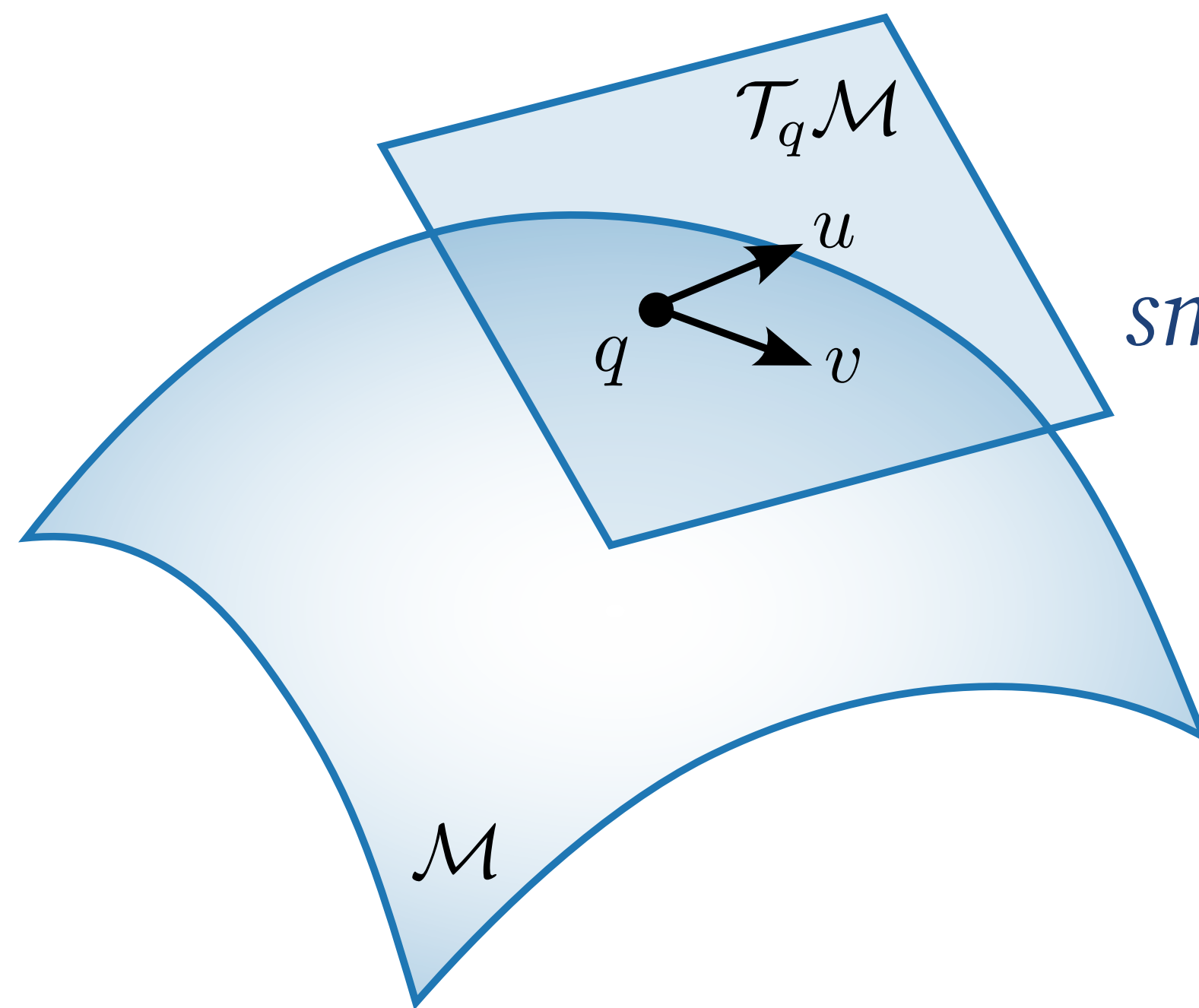
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A Riemannian metric attaches a **notion of distance** to the manifold
— generalizing both the isotropic and anisotropic cases

Euclidean inner product: $\langle u, v \rangle = u^T v = u^T I v$



smoothly varying inner product between tangent vectors
 $u, v \in \mathcal{T}_q \mathcal{M}$

$$\langle u, v \rangle_q = u^T \underbrace{G(q)}_{\text{symmetric, positive-definite matrix}} v$$

symmetric, positive-definite matrix